

EXAMPLE 15: Balance the equation:



H and Cl are not balanced, hence:



EXERCISE 31 Balance the following equations.

1. $\text{Na} + \text{Cl}_2 \longrightarrow \text{NaCl}$
2. $\text{CO} + \text{O}_2 \longrightarrow \text{CO}_2$
3. $\text{Al} + \text{O}_2 \longrightarrow \text{Al}_2\text{O}_3$
4. $\text{HgO} \longrightarrow \text{Hg} + \text{O}_2$
5. $\text{Mg} + \text{N}_2 \longrightarrow \text{Mg}_3\text{N}_2$
6. $\text{P}_4 + \text{O}_2 \longrightarrow \text{P}_4\text{O}_{10}$
7. $\text{Al} + \text{F}_2 \longrightarrow \text{AlF}_3$
8. $\text{PbO} + \text{C} \longrightarrow \text{Pb} + \text{CO}_2$
9. $\text{CH}_4 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$
10. $\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
11. $\text{Ca} + \text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 + \text{H}_2$
12. $\text{Fe} + \text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2$
13. $\text{MgCO}_3 + \text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
14. $\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
15. $\text{Mg} + \text{AgNO}_3 \longrightarrow \text{Mg(NO}_3)_2 + \text{Ag}$

EXERCISE 32 Write names for the elements and compounds in the equations for Exercise 31.

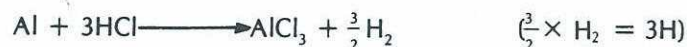
EXAMPLE 16: Balance the equation:



Both the H and Cl are not balanced. Balance the Cl first since H_2 contains no other elements, enabling H to be balanced last.



Balancing the H



The above equation is satisfactory but the fraction can be eliminated by multiplying by 2



EXAMPLE 17: Balance the equation:



NH_4^+ and NO_3^- ions appear on both sides of the equation. It is easier to balance the equation by balancing these groups, rather than dividing them into the elements of which they are made.

Neglecting these groups, C and O are balanced, but not H.

Balance the H in HNO_3 .



Simultaneously balance NH_4^+ and NO_3^-



EXERCISE 33 Balance the following equations.

1. $\text{H}_2\text{S} + \text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{SO}_2$
2. $\text{Al} + \text{AgNO}_3 \longrightarrow \text{Al(NO}_3)_3 + \text{Ag}$
3. $\text{NaOH} + \text{H}_3\text{PO}_4 \longrightarrow \text{Na}_3\text{PO}_4 + \text{H}_2\text{O}$
4. $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$
5. $\text{NaOH} + \text{Fe(NO}_3)_2 \longrightarrow \text{Fe(OH)}_2 + \text{NaNO}_3$
6. $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow \text{Fe} + \text{CO}_2$
7. $\text{Cu} + \text{HNO}_3 \longrightarrow \text{Cu(NO}_3)_2 + \text{H}_2\text{O} + \text{NO}_2$
8. $\text{NH}_3 + \text{CuO} \longrightarrow \text{Cu} + \text{H}_2\text{O} + \text{N}_2$
9. $\text{Al}_2(\text{CO}_3)_3 + \text{HCl} \longrightarrow \text{AlCl}_3 + \text{H}_2\text{O} + \text{CO}_2$
10. $\text{FeS} + \text{O}_2 \longrightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$

EXERCISE 34 Write names for the elements and compounds in the equations for Exercise 33.

EXAMPLE 18: Balance the equation:



H and Cl are unbalanced, hence:



EXAMPLE 19: Balance the equation:

Calcium hydroxide + phosphoric acid $\longrightarrow$ calcium phosphate + water



Balance Ca



Balance PO_4^{3-}



Balance H and O (other than that in PO_4^{3-})

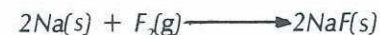


EXERCISE 35 Write fully balanced equations for the following.

1. hydrogen + oxygen $\longrightarrow$ water
2. carbon + oxygen $\longrightarrow$ carbon dioxide
3. sodium + chlorine $\longrightarrow$ sodium chloride
4. potassium + oxygen $\longrightarrow$ potassium oxide
5. magnesium + oxygen $\longrightarrow$ magnesium oxide
6. magnesium + hydrochloric acid $\longrightarrow$ magnesium chloride + hydrogen
7. ammonia + water $\longrightarrow$ ammonium hydroxide
8. zinc + hydrochloric acid $\longrightarrow$ zinc chloride + hydrogen
9. sodium + sulfuric acid $\longrightarrow$ sodium sulfate + hydrogen
10. calcium carbonate + hydrochloric acid $\longrightarrow$ calcium chloride + water + carbon dioxide
11. hydrogen + nitrogen $\longrightarrow$ ammonia
12. aluminium + oxygen $\longrightarrow$ aluminium oxide
13. calcium oxide + water $\longrightarrow$ calcium hydroxide
14. calcium hydroxide + carbon dioxide $\longrightarrow$ calcium carbonate + water
15. ammonia + sulfuric acid $\longrightarrow$ ammonium sulfate
16. silver nitrate + magnesium chloride $\longrightarrow$ silver chloride + magnesium nitrate
17. potassium nitrate $\longrightarrow$ potassium nitrite + oxygen
18. sodium hydroxide + carbon dioxide $\longrightarrow$ sodium carbonate + water
19. potassium bicarbonate $\longrightarrow$ potassium carbonate + water + carbon dioxide
20. zinc + silver nitrate $\longrightarrow$ zinc nitrate + silver

EQUATIONS SHOW THE RATIO IN WHICH MOLECULES, ATOMS OR IONS REACT.

EXAMPLE 20: What information is given by the equation below?



For every two atoms of sodium in a metallic lattice that react,

- one molecule of fluorine reacts
- two sodium ions and two fluoride ions in an ionic lattice are formed.

EXERCISE 36 What information is given by each of the equations below?

1. $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{H}_2\text{O(g)}$
2. $\text{C(s)} + \text{O}_2\text{(g)} \longrightarrow \text{CO}_2\text{(g)}$
3. $2\text{KHCO}_3\text{(s)} \longrightarrow \text{K}_2\text{CO}_3\text{(s)} + \text{H}_2\text{O(g)} + \text{CO}_2\text{(g)}$
4. $\text{Zn(s)} + 2\text{AgNO}_3\text{(aq)} \longrightarrow \text{Zn(NO}_3)_2\text{(aq)} + 2\text{Ag(s)}$
5. $3\text{H}_2\text{(g)} + \text{N}_2\text{(g)} \longrightarrow 2\text{NH}_3\text{(g)}$